

CRITICAL EQUATIONS NOTES

EDWARD CHERNYSH

1. SETTING UP THE PROBLEM

Let $\Omega \subset \mathbb{R}^n$ be a non-empty bounded domain, containing the origin, such that $\partial\Omega$ is smooth. Let $g : \Omega \times \mathbb{R} \rightarrow \mathbb{R}$ be a Carathéodory function; that is, $g(x, \cdot)$ is continuous for almost every $x \in \Omega$ and $g(\cdot, u)$ is measurable for all $u \in \mathbb{R}$. Also, we require that g satisfy the bound

$$|g(x, u)| \leq C (1 + |u|^{q'-1}) \quad (1.1)$$

for all $(x, u) \in \Omega \times \mathbb{R}$ and a constant $C > 0$ independent of x, u . Here, we assume that $q' \in (1, q)$ where q is given by

$$q := \frac{np}{n - p(1 + a - b)}$$

with

$$1 < p < n, \quad a < \frac{n-p}{p}, \quad \text{and} \quad a \leq b < a + 1. \quad (1.2)$$

We will be interested in the following critical equation of the Caffarelli-Kohn-Nirenberg type:

$$\begin{cases} -\operatorname{div} (h(x) |\nabla u|^{p-2} \nabla u) = f(x) |u|^{q-2} u + g(x, u) & \text{in } \Omega, \\ u \in \mathcal{D}^{1,p}(\Omega, h). \end{cases} \quad (1.3)$$

Here, $\mathcal{D}^{1,p}(\Omega, h)$ is the completion of $C_c^\infty(\Omega)$ with respect to the norm

$$\|u\|_{\mathcal{D}^{1,p}(\Omega, h)} := \|\nabla u\|_{L^p(\Omega, h)} = \left(\int_{\Omega} h(x) |\nabla u(x)|^p \, dx \right)^{1/p}.$$

Hence, $\mathcal{D}^{1,p}(\Omega, h)$ is precisely the weighted Sobolev space $W_0^{1,p}(\Omega, h)$. We will also be interested in the weighted homogeneous Sobolev space $\mathcal{D}^{1,p}(\mathbb{R}^n, h)$ which is defined to be the completion of $C_c^\infty(\mathbb{R}^n)$ with respect to the norm

$$\|u\|_{\mathcal{D}^{1,p}(\mathbb{R}^n, h)} := \left(\int_{\mathbb{R}^n} h(x) |\nabla u(x)|^p \, dx \right)^{1/p}.$$

In the context of Problem (1.3), we will always assume that the following two conditions are satisfied:

Date: May 11, 2019.

- (1) $f(x) = |x|^{-bq} \hat{f}(x)$, where $\hat{f}$ is continuous on the closure of Ω ;
- (2) $h(x) = |x|^{-ap} \hat{h}(x)$, with $\hat{h}$ both continuous and positive on $\overline{\Omega}$. We then extend $\hat{h}$ to all of $\mathbb{R}^n$ by setting $\hat{h}(x) = 0$ for $x \notin \overline{\Omega}$. In particular, h becomes a well defined weight.

1.1. Discussing Weighted Lebesgue Spaces. In this subsection, we relax any assumption on p and assume that $p \in [1, \infty)$ is arbitrary, but fixed. Let $(X, \mathfrak{M}, \mu)$ be a measure space. A measurable function $\omega : X \rightarrow (0, \infty]$ is said to be a *weight* on the measurable space $(X, \mathfrak{M})$. We then define the weighted Lebesgue space to be the set of all measurable functions $f : X \rightarrow \mathbb{R}$ such that

$$\int_X |f|^p \omega \, d\mu < \infty.$$

The space of all such functions is denoted by $L^p(X, \mu; \omega)$ and can be made into a normed space by attaching the norm

$$\|f\|_{L^p(X, \mu; \omega)} := \left(\int_X |f|^p \omega \, d\mu \right)^{1/p}.$$

We now claim that there is nothing special about these weighted Lebesgue spaces. In fact, we shall show that they are identical to $L^p(X, \nu)$ for a suitable measure ν on the measurable space $(X, \mathfrak{M})$ that has the same measure zero sets as μ . Indeed, consider the set function $\nu : \mathfrak{M} \rightarrow [0, \infty]$ given by

$$\nu(E) := \int_E \omega \, d\mu, \quad E \in \mathfrak{M}.$$

Clearly, $\nu(\emptyset) = 0$. If $\{E_k\}_{k \geq 1}$ is a countable disjoint family from $\mathfrak{M}$, then the monotone convergence theorem ensures that, for $E := \bigcup_{k=1}^{\infty} E_k$, there holds

$$\begin{aligned} \nu(E) &= \int_{\bigcup_{k=1}^{\infty} E_k} \omega \, d\mu = \lim_{l \rightarrow \infty} \int_{\bigcup_{k=1}^l E_k} \omega \, d\mu = \lim_{l \rightarrow \infty} \sum_{k=1}^l \nu(E_k) \\ &= \sum_{k=1}^{\infty} \nu(E_k). \end{aligned}$$

Hence, ν is a measure on $(X, \mathfrak{M})$. Our assertion then becomes the following:

Proposition 1.1. *Let $(X, \mathfrak{M}, \mu)$ be a measure space and let ω be a weight on $(X, \mathfrak{M})$. Assume that $1 \leq p < \infty$. With ν given as above, $L^p(X, \mu; \omega)$ and $L^p(X, \nu)$ are identical. In particular, one has*

$$\left(\int_X |f|^p \omega \, d\mu \right)^{1/p} = \left(\int_X |f|^p \, d\nu \right)^{1/p} \quad (1.4)$$

for all f belonging to either space. Moreover, μ and ν have the same null sets.

Proof. Since ω is positive, it is clear that μ and ν share the same null sets. Consequently, it would be enough to show that (1.4) holds for *all* measurable functions f . Now, let (φ_k) be a sequence of non-negative simple functions such that $\varphi_k(x) \nearrow |f(x)|^p$ for every $x \in X$. For each $k \geq 1$, we write

$$\varphi_k(x) = \sum_{j=1}^{N_k} \alpha_{j,k} \mathbf{1}_{E_{j,k}}(x), \quad \alpha_{j,k} \geq 0 \text{ and } E_{j,k} \in \mathfrak{M}.$$

An application of the monotone convergence theorem then asserts that

$$\int_X |f|^p d\nu = \lim_{k \rightarrow \infty} \int_X \varphi_k(x) d\nu$$

where, for each $k \in \mathbb{N}$,

$$\begin{aligned} \int_X \varphi_k(x) d\nu &= \sum_{j=1}^{N_k} \alpha_{j,k} \nu(E_{j,k}) = \sum_{j=1}^{N_k} \alpha_{j,k} \int_X \mathbf{1}_{E_{j,k}}(x) \omega(x) d\mu \\ &= \int_X \varphi_k(x) \omega(x) d\mu. \end{aligned}$$

Therefore, invoking the monotone convergence theorem a final time,

$$\int_X |f|^p d\nu = \lim_{k \rightarrow \infty} \int_X \varphi_k(x) \omega(x) d\mu = \int_X |f|^p \omega d\mu.$$

This completes the proof. $\square$

This proposition shows that the familiar results about plain old L^p -spaces also applies to these weighted L^p -spaces. Essentially, weights are only nice ways of constructing exotic measures using well understood pre-existing ones (e.g. the Lebesgue measure).

1.2. Useful inequalities and observations. We also recall the Caffarelli-Kohn-Nirenberg (CKN) inequality which asserts that, for a suitable constant $C = C(n, a, b, p) > 0$, one has

$$\left(\int_{\mathbb{R}^n} |x|^{-bq} |u(x)|^q dx \right)^{1/q} \leq C \left(\int_{\mathbb{R}^n} |x|^{-ap} |\nabla u(x)|^p dx \right)^{1/p} \quad (\text{CKN})$$

for every $u \in C_c^\infty(\mathbb{R}^n)$. In particular, this gives an embedding $\mathcal{D}^{1,p}(U, |x|^{-ap}) \hookrightarrow L^q(U, |x|^{-bq})$ for every open set $U \subseteq \mathbb{R}^n$. Therefore, our next question is when this embedding is compact? This begins with a glimpse into the *Muckenhoupt* weight classes.

1.2.1. Muckenhoupt weights.